



Innovative Applications of O.R.

Fighting sampling bias: A framework for training and evaluating credit scoring models

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ARTICLE INFO

Keywords:

OR in banking
Machine learning
Credit scoring
Sampling bias
Reject inference

ABSTRACT

Scoring models support decision-making in financial institutions. Their estimation and evaluation rely on labeled data from previously accepted clients. Ignoring rejected applicants with unknown repayment behavior introduces sampling bias, as the available labeled data only partially represents the population of potential borrowers. This paper examines the impact of sampling bias and introduces new methods to mitigate its adverse effect. First, we develop a bias-aware self-labeling algorithm for scorecard training, which debiases the training data by adding selected rejects with an inferred label. Second, we propose a Bayesian framework to address sampling bias in scorecard evaluation. To provide reliable projections of future scorecard performance, we include rejected clients with random pseudo-labels in the test set and use Monte Carlo sampling to estimate the scorecard's expected performance across label realizations. We conduct extensive experiments using both synthetic and observational data. The observational data includes an unbiased sample of applicants accepted without scoring, representing the true borrower population and facilitating a realistic assessment of reject inference techniques. The results show that our methods outperform established benchmarks in predictive accuracy and profitability. Additional sensitivity analysis clarifies the conditions under which they are most effective. Comparing the relative effectiveness of addressing sampling bias during scorecard training versus evaluation, we find the latter much more promising. For example, we estimate the expected return per dollar issued to increase by up to 2.07 and up to 5.76 percentage points when using bias-aware self-labeling and Bayesian evaluation, respectively.

1. Introduction

Assessing and mitigating the risk of money lending are crucial management tasks in the credit business. Machine learning (ML)-based scoring models support corresponding tasks by estimating credit default probabilities. Such estimates guide, for example, loan approval decisions, loss provisioning, and regulatory capital calculations. The paper studies the sampling bias problem, which routinely arises in the lending business and harms the quality of scorecard-based decisions.

To illustrate the problem, consider a loan applicant who seeks financing. The financial institution (FI) predicts the applicant's probability of repayment and decides the application accordingly. The FI can

observe the repayment behavior of accepted clients and eventually determine if the debt was repaid. We refer to the groups of applicants who repay and default as *good* and *bad* risks, respectively. Sampling bias is a missing data problem. The FI cannot observe the repayment behavior of rejected clients (Wu & Hand, 2007). More specifically, developing scorecards requires *labeled* data with known outcomes (i.e., observed repayment status). Lacking labels for rejects, the FI uses data from previously accepted clients for scorecard development. This sample differs from the borrower population, which implies a bias (Banasik et al., 2003).

Credit scorecards impact borrowers, risk management, firm profitability, and, ultimately, society. This warrants concern about sampling

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bias. For example, in 2023, the total outstanding amount of consumer credit in the US exceeded \$5,019 billion (The Federal Reserve, 2023). Scorecards have played an important role in the approval of this amount. This shows how scorecards determine access to finance, which is a crucial factor for economic (in)equality (Wei et al., 2016). Beyond retail, scorecards also inform corporate lending (Korangi et al., 2023), which emphasizes their prevalence and the potential implications of sampling bias.

The objectives of this paper are to clarify the effect of sampling bias on scorecard predictions and to develop methodologies that mitigate its adverse effect. In pursuit of this objective, the paper provides two methodological contributions. First, we introduce bias-aware self-labeling (BASL), a reject inference (RI) framework that extends our previous work in Kozodoi et al. (2019) and addresses sampling bias during the training of a scoring model. Unlike existing data augmentation methods including our previous work, we design BASL to prioritize prediction performance over debiasing and show this strategy to produce better predictions.

A crucial step in the scorecard lifecycle concerns model evaluation. Estimating scorecard accuracy requires labeled data, which again limits the data an FI can use for evaluation to the (biased) sample of accepted clients. Therefore, our second methodological contribution consists of a Bayesian evaluation framework (BE) that leverages the unlabeled data of rejected clients and estimates scorecard accuracy from a joint sample of accepted and rejected clients, which reflects the borrower population. BE comprises techniques to sample labels for rejects and facilitates a reliable scorecard assessment under sampling bias.

Beyond novel methodology, the paper contributes original insights into the severity of the sampling bias problem in credit scoring and the efficacy of bias mitigation strategies from extensive empirical experimentation using simulations and real-world micro-lending data. While the simulation study facilitates testing novel methodology in a controlled environment and sensitivity analysis to identify boundary conditions, a distinct feature of our real-world data is that it comprises an unbiased sample, which the lender collected by accepting all incoming applications for a short period. This sample mimics the setting of Bravo et al. (2013), who also employed a data set without rejected loans for their study on credit scoring systems for managing loans for micro-entrepreneurs. In the context of RI, few previous studies, including seminal work of Banasik and Crook (2007), employed unbiased data. The prevailing approach in prior work is to test RI methodology using (biased) data from accepted applications. Our empirical analysis shows this approach produces misleading results. Being able to overcome this problem emphasizes the empirical contribution of our study.

We proceed with discussing relevant background (Section 2) and related work (Section 3) before presenting the novel methodologies for scorecard training (Section 5) and evaluation (Section 4). Next, we elaborate on our experimental design (Section 6), report empirical results (Section 7), and discuss their implications (Section 8). Section 9 concludes the paper. The online appendix³ provides auxiliary results complementing the main paper.

2. Theoretical background and problem setting

This section formalizes sampling bias and the RI problem. Let $X \in \mathbb{R}^k$ denote a loan applicant. The matrix of the applicants' attributes is denoted as $\mathbf{X} = (X_1, \dots, X_n)^\top$, and $\mathbf{y} = (y_1, \dots, y_n)^\top$ is a random vector of binary labels, indicating if the applicant repays the loan ($y = 0$) or defaults ($y = 1$). Suppose $\mathbf{X}$ and $\mathbf{y}$ have marginal distributions and a joint distribution denoted by $\mathbf{P}_X$, $\mathbf{P}_Y$, and $\mathbf{P}_{XY}$. Given a set of independent and identically distributed applications $D = \{(\mathbf{X}, \mathbf{y})\}$

with $(\mathbf{X}, \mathbf{y}) \sim \mathbf{P}_{XY}$, a financial institution uses a scorecard $f(X)$ that approximates $\mathbf{P}(y = 1|X)$ to split D into two subsets: accepts D^a and rejects D^r , $D = D^a \sqcup D^r$. The repayment behavior is eventually observed for D^a . The labels of rejects remain unknown. Thus, D exhibits missingness with respect to repayment outcomes y . Binary variable $a \in \{0, 1\}$ indicates if the applicant's repayment outcome is observed ($a = 1$) or missing ($a = 0$), which corresponds to whether the applicant was accepted.

The missing data literature distinguishes three types of missingness (Little & Rubin, 2019). Labels are missing completely at random (MCAR) if $\mathbf{P}(a|X, y) = \mathbf{P}(a)$. Labels are missing at random (MAR) if $\mathbf{P}(a|X, y) = \mathbf{P}(a|X)$, implying that label missingness depends on the applicants' attributes but not on the repayment status. When labels are missing not at random (MNAR), the missingness also depends on y due to unobserved factors, which cannot be explained by the attributes. Formally, the data exhibits MNAR if $\mathbf{P}(a|X, y) \neq \mathbf{P}(a|X)$.

In credit scoring, we can ignore MCAR, which corresponds to a random acceptance policy. Approving loans using a scorecard causes D^a to have different empirical distributions compared to $\mathbf{P}_{XY}$, $\mathbf{P}_X$ and $\mathbf{P}_Y$ and creates sampling bias. In practice, it is difficult to distinguish MNAR and MAR because the unobserved factors are not accessible. MAR occurs if a FI does not use external information apart from $\mathbf{X}$ to make acceptance decisions (e.g., always relies on predictions of the same scorecard). If scorecards and their underlying data change over time, we can still assume MAR if all changes have been documented and the policy by which loans were approved is completely known. On the other hand, cases like early repayment of loan obligations lead to censoring and, in turn, confounding, which implies MNAR. Another cause of MNAR is overwriting of scorecard recommendations based on attributes not included in $\mathbf{X}$. For example, applicants with a County Court Judgment may be manually rejected even if the scorecard prediction is positive (Banasik & Crook, 2007). Such actions tie label missingness to factors unknown to the model $f(X)$. MNAR can also occur when some of the features in $\mathbf{X}$ included in a previous scorecard can no longer be used by a FI (e.g., due to new data privacy regulations or changes in data providers).

Prior work provides several approaches for missing value imputation under MAR and MNAR to correct sampling bias (Little & Rubin, 2019). However, full debiasing might not be the most suitable strategy in a credit scoring context. Facing a prediction problem, the primary goal of addressing sampling bias in the training data $D^a = \{(\mathbf{X}^a, \mathbf{y}^a)\}$ is to facilitate accurate approximation of $\mathbf{P}(y = 1|X)$. Full debiasing of D^a achieves this goal but may be difficult to achieve; especially under MNAR and/or when a large number of features complicate density estimation, which many imputation techniques entail (Peng et al., 2023). Full debiasing may also be dispensable in the pursuit of higher predictive accuracy. For example, under MAR, posterior probability models such as logistic regression do not require bias correction (Banasik & Crook, 2007). More generally, given that the true missingness mechanism is unknown in practice, approaches for debiasing the training data can introduce noise. This diminishes the beneficial effect of reducing bias by introducing variance, leaving the net effect on the quality of the approximation of $\mathbf{P}(y = 1|X)$, that is scorecard accuracy, unclear. We examine the relationship of sampling bias and scorecard accuracy in Appendix C.3 and C.4, and evidence a trade-off. A small reduction of bias can unlock sizeable gains in scorecard performance whereas over-emphasizing debiasing produces less accurate predictions.

Apart from affecting scorecard training, sampling bias also impedes model evaluation under MAR and MNAR. A holdout set H^a drawn from D^a is not representative of $D \sim \mathbf{P}_{XY}$ because D^a contains applications predicted as least risky. Thus, evaluating $f(X)$ using the data of accepts will overestimate scorecard performance (Banasik et al., 2003).

Consolidating the theoretical considerations, we formalize our research goals as follows: given a set of labeled accepts D^a and unlabeled rejects D^r , we aim to: (i) infer a function $f(X)$ that approximates $\mathbf{P}(y = 1|X)$ and generalizes well over applications from $\mathbf{P}_{XY}$ and (ii)

³ URL will be available in the final paper. All references to appendices refer to the online appendix.

estimate the predictive performance of $f(X)$ over applications from $\mathbf{P}_{XY}$. Exploiting the information in D' , we aim to address sampling bias in both tasks and improve, respectively, scorecard performance and the accuracy of predictions of scorecard performance. Facing uncertainty related to the type of missingness, we perform sensitivity analysis to test our propositions in MAR and MNAR settings.

3. Related work

Sampling bias is extensively studied in different streams of literature with different perspectives. Examples include work on missing data problems due to non-random sampling (e.g., Peng et al., 2023) or on policy learning (e.g., Athey & Wager, 2021). Further, a rich literature on sampling bias and RI in credit scoring exists (e.g., Anderson et al., 2023; Banasik & Crook, 2007; Hand & Henley, 1993). Facing a large and diverse set of prior work, this section sketches related approaches in domains different from credit risk while also elaborating on relationships with prior RI studies and how we contribute to the field. We augment the discussion with a detailed tabular analysis of relevant papers in Appendix A of the online companion, where Table A.1 concentrates on sampling bias correction in general and Table A.2 focuses on RI studies in credit scoring.

3.1. Training under sampling bias

Representation change is a family of bias correction methods applied in the data preprocessing stage before training a corrected model. They assume MAR and use a mapping function Φ to project features into a new representational space $\mathbf{Z}$, $\Phi : \mathbf{X} \rightarrow \mathbf{Z}$, such that the training data distribution over $\mathbf{Z}$ is less biased and $\Phi(X)$ retains as much information about X as possible. A suitable representation is found by maximizing a distribution similarity measure such as the distance between the distribution moments in a kernel space (Borgwardt et al., 2006). Recently, generative models have gained popularity in learning feature transformations. For example, Atan et al. (2018) train a deep autoencoder with a mismatch penalty and extract the corrected data representation from the bottleneck layer. A potential disadvantage for credit scoring concerns interpretability. Regulatory compliance requires FIs to ensure comprehensible scoring models, which is difficult when employing transformed feature sets.

Model-based bias correction methods adjust a learning algorithm to account for sampling bias. In his pioneering work, Heckman (1979) proposed a two-stage least-squares model for the MNAR setup. This model simultaneously estimates an outcome equation and a sample selection process, which facilitates eliminating bias in the estimated model parameters to yield consistent estimates. Meng and Schmidt (1985) developed a bivariate probit model with non-random sample selection for setups where the outcome variable is binary. Their model represents a sound approach for credit scoring under assumptions of MNAR and normally distributed residuals in the estimated equations. Notably, the work of Heckman (1979) and Meng and Schmidt (1985) exemplifies the focus in much prior work on missing data in that they emphasize unbiasedness in the estimates of model parameters. Many ML models are nonparametric and do not explicate internal parameters. Thus, our main interest is the impact of sampling bias on model predictions and their evaluation. We consider the bivariate probit models as a benchmark in our empirical analysis (see Section 7.2) and, as a byproduct of our research, suggest ways to examine bias in ML models analogously to how it is done in linear models using explainable AI tools (see Appendix D.2).

Another research stream considers mixture models for bias correction (e.g., Feelders, 1999). These operate under the MAR assumption and treat the data as drawn from a mixture of two distributions: training and population. Learning from both samples, mixture models infer labels of new examples using the conditional expectation-maximization algorithm. A disadvantage of model-based methods is

that they are embedded in a learning algorithm, which requires a specific classifier. Much prior work has focused on (semi-)parametric models with particular assumptions. Yet, there is evidence that other non-parametric approaches, such as XGB, perform better in credit scoring (e.g., Gunnarsson et al., 2021).

Reweighting is another method that rebalances the training loss towards representative examples. Weights of the training examples, known as importance weights or propensity scores, can be computed as a ratio of two distribution densities: $w(X) = p_D(X)/p_{D^a}(X)$. High values of $w(X)$ indicate that X is more likely drawn from $\mathbf{P}_{XY}$ and is, therefore, more important for training. Importance weights can be estimated in several ways (e.g., Huang et al., 2006) and employed during training using, e.g., weighted least squares. Since reweighting relies on attributes in $\mathbf{X}$, it assumes MAR and cannot correct for MNAR. For example, a reweighted training set still consists of previously accepted clients and misses certain distribution regions populated by rejects only. However, reweighting can still be helpful under MNAR as it may reduce errors in estimating a model from the training sample (Banasik & Crook, 2007).

The credit scoring literature has also explored the idea of data augmentation — expanding the training sample by labeling and appending examples from D' . The augmented sample covers a wider distribution region, which may reduce sampling bias. Prior work suggests different approaches that use a model trained over D^a to label rejects. A classic example is hard cutoff augmentation (HCA), which labels rejects by comparing their predicted scores with the accepts-based model to a predefined threshold. Under sampling bias, reliance on the accepts-based model may increase the risk of error propagation when labeling rejects. Extrapolating predictions of the accepts-based scorecard on rejects is a valid technique for posterior probability classifiers under MAR but suffers from the omitted variable bias under MNAR (Banasik et al., 2003).

Parceling aims to improve upon HCA by considering rejects as riskier than accepts. Parceling splits rejects into segments based on the predicted score range, and labels rejects within each range proportional to the assumed PD. A decision-maker can alter the probabilities compared to the ones observed within the same score range on D^a . Thus, parceling can work under MNAR if the decision-maker can correctly specify the change in the PDs across the groups of applicants.

3.2. Evaluation under sampling bias

Sampling bias also affects model evaluation. An estimate of model performance derived from a biased sample may not generalize to unseen data, which impedes standard model evaluation strategies such as cross-validation (Sugiyama et al., 2007). To address this problem, prior work proposes generalization error measures, whose asymptotic unbiasedness is maintained under sampling bias. This includes the modified Akaike information criterion (MAIC, Shimodaira, 2000) and the generalization error measure suggested by Sugiyama and Müller (2006). Both measures rely on density ratio estimation, which may be difficult in higher dimensions. Measures like the MAIC are also limited to parametric learners.

Evaluation under sampling bias is also studied in off-policy evaluation research, which focuses on the evaluation of a policy (e.g., a classifier) in a contextual bandit setting with incomplete historical data (Athey & Wager, 2021). In this setup, a policy reward depends on the action of a decision-maker and is only partially observed. A prominent policy evaluation method is importance-weighted validation, which reweights the reward towards more representative examples in the evaluation set using importance weights (Sugiyama et al., 2007).

Reweighting produces biased estimates if the past policy is modeled incorrectly (Dudík et al., 2014). In our context, this implies that the attributes $\mathbf{X}$ do not explain prior acceptance decisions accurately, and the data exhibits MNAR. Dudík et al. (2014) recommend doubly robust

(DR) estimators, which combine estimating importance weights with predicting policy reward (i.e., classifier loss). DR produces unbiased estimates if at least one of the modeled equations is correct. However, using DR in credit scoring is difficult. The contextual bandit setting considers a set of actions and assumes we observe a reward for the chosen action. DR can then impute the reward for other actions. In credit scoring, however, we do not observe a reward for rejected clients, which complicates the imputation of reward substantially. Also, measuring reward as classifier loss limits DR to performance measures calculated on the level of an individual loan. This prohibits using DR with popular rank-based metrics, such as the area under the ROC curve (AUC).

3.3. Applications in credit scoring

Sampling bias has received much attention in credit scoring. Prior work focuses on scorecard training and tests debiasing methods, including the Heckman model (e.g., Banasik et al., 2003) and data augmentation techniques such as HCA (e.g., Crook & Banasik, 2004). A commonly used reweighting approach is banded weights, which uses the bands of predicted probabilities of default to form clusters (Banasik & Crook, 2005). Maldonado and Paredes (2010) were the first to propose semi-supervised learning for RI. Since then, semi-supervised learning has become the most popular approach to tackling sampling bias in loan approval, as shown in the survey of the empirical RI literature in Table A.2 of Appendix A.

Evaluation under sampling bias has received less attention in the credit risk literature. Using a proprietary data set with labeled rejected applicants, Banasik et al. (2003) illustrate discrepancies between the accuracy estimates on a sample from accepts and a representative holdout sample and use the latter to judge the value of RI. To our knowledge, prior work has not considered techniques that mitigate the impact of sampling bias on model evaluation in the absence of such a sample.

A common characteristic of RI studies is that they use low-dimensional data (see Table A.2). Recent studies emphasize the merit of alternative data such as applicants' digital footprints (e.g. Djeundje et al., 2021) or networks (Óskarsdóttir & Bravo, 2021). Further, few RI studies express the gains from RI in terms of profitability or use a proper representative sample to assess the bias correction benefits; exceptions include Banasik and Crook (2005), Chen and Astebro (2001). We strive to address these limitations by using high-dimensional FinTech data, evaluating performance on a representative sample from the borrowers' population, and examining the business impact of RI.

4. Bayesian evaluation framework

Estimating scorecard performance is crucial. Policy changes such as purchasing additional data about applicants or changing the acceptance rate entail comparing scorecards' performance. Evaluating a scorecard on a biased validation set of accepted applicants produces biased performance estimates. The true scorecard performance in production will not match the expectations raised during evaluation, which may lead to overestimating its business value and suboptimal policy decisions. This section introduces the Bayesian evaluation framework (BE) that aims to mitigate the adverse effect of sampling bias on model evaluation.

The performance prediction task can be described as follows: given a population of loan applicants $D = D^a \cup D^r$, we aim to evaluate the scoring model $f(X)$ on a representative holdout set denoted as H , $H \subset D$ to assess its true ability to infer the creditworthiness of new applicants. Standard performance measures require knowledge of labels for all cases in H . In practice, labels in $H^a = H \cap D^a$ are known but labels of rejects in $H^r = H \cap D^r$ are missing. Thus, banks commonly assess $f(X)$ on H^a (i.e., evaluate based on accepts). Empirical results in Section 7 illustrate the inadequacy of this approach and motivate our proposition.

Specifically, BE extends standard performance metrics by adding unlabeled rejects to the evaluation set. Assume $f(X)$ is evaluated on H using an arbitrary metric $M(f, H, \tau)$, where τ is a vector of metric meta-parameters (e.g., PD cutoff). Algorithm 1 computes the Bayesian extension of M denoted as $BM(f, H, \tau)$. Since labels in H^r are unknown, we use a prior of the label distribution among rejects $P(y^r, X^r)$ to assign random pseudo-labels. We then assess $f(X)$ on a representative sample of labeled accepts and pseudo-labeled rejects.

BE uses Monte Carlo sampling. Each unknown label is drawn from a binomial distribution with the probability set to the prior for that rejected example. The Bayesian extension of a metric is computed by averaging the metric across label realizations. The sampling iterations are terminated once the incremental change of the average does not exceed a convergence threshold ϵ .

The accuracy of the estimated metric depends on the choice of the prior, which governs the sampling of class labels of rejects. We propose leveraging the attributes of examples in D^r denoted by X^r and estimate the prior $P(y^r|X^r)$ by rescoreing rejects with a model that has been used to support loan approval decisions in the past. The proposed framework stands on the trade-off of two components: benefits from evaluating a model on a more representative validation sample and additional noise in the labels of rejects based on an imperfect prior. We establish through sensitivity analysis (see Section 7.1) that gains from extending the evaluation sample typically outweigh losses due to noise, which facilitates more accurate performance predictions.

It is useful to assess credit scorecards from different angles using complementary performance measures (Lessmann et al., 2015). In this regard, a notable feature of BE is that it supports arbitrary performance measures. For example, characterizing the operating conditions of a scorecard, such as class priors and misclassification costs, loan characteristics (e.g., principal, interest rate), or estimates of the LGD and EAD can be incorporated in the parameter vector τ to approximate scorecards' profitability (Verbraken et al., 2014).

```

input : model  $f(X)$ , evaluation set  $H$  with labeled accepts
          $H^a = \{(X^a, y^a)\}$  and unlabeled rejects  $H^r = \{X^r\}$ , prior
          $P(y^r|X^r)$ , metric  $M(f, H, \tau)$ , meta-parameters  $j_{max}, \epsilon$ 
output: Bayesian evaluation metric  $BM(f, H, \tau)$ 
1  $j = 0; \Delta = \epsilon; E = \{\}$ ; // initialization
2 while ( $j \leq j_{max}$ ) and ( $\Delta \geq \epsilon$ ) do
3    $j = j + 1$ 
4    $y^r = \text{binomial}(1, P(y^r|X^r))$ ; // generate labels of rejects
5    $H_j = \{(X^a, y^a)\} \cup \{(X^r, y^r)\}$ ; // construct evaluation
   sample
6    $E_j = \frac{1}{j} \sum_{i=1}^j M(f, H_i, \tau)$ ; // evaluate  $f(X)$ 
7    $\Delta = E_j - E_{j-1}$ ; // check metric convergence
8 end
return:  $BM(f, H, \tau) = E_j$ 

```

Algorithm 1: Bayesian Evaluation Framework

5. Bias-aware self-labeling framework

The BASL framework is based on self-labeling (also known as self-training), a semi-supervised learning algorithm that uses unlabeled examples to improve predictive performance (Levatić et al., 2017). Given a labeled set D^a and an unlabeled set D^r , self-labeling trains a supervised model over D^a . This model scores examples in D^r . If predictions exceed a specified threshold, examples are assigned the corresponding labels and appended to D^a . The model is then retrained on D^a and used to score the remaining examples in D^r . This repeats until a stopping criterion is met.

Sampling bias impedes the effectiveness of self-labeling. First, labeling rejects based on the confidence of the accepts-based model may be misleading. Rejects come from a different distribution region, and the model can produce overconfident predictions, which become less reliable as the difference between the two samples increases. These errors are propagated through consecutive labeling iterations. Second,

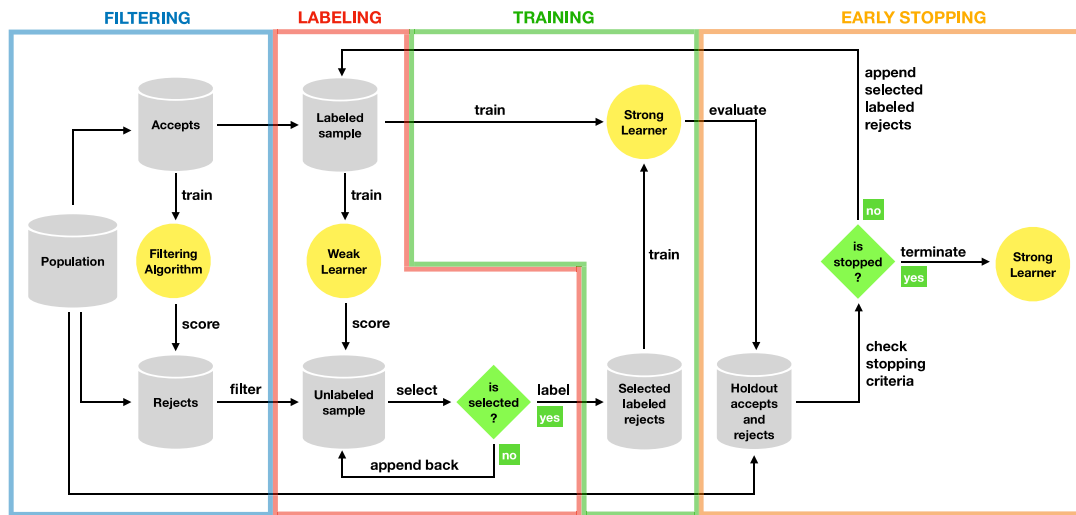


Fig. 1. Bias-aware self-labeling framework.

Notes: The filtering algorithm (isolation forest) filters out rejects before the self-labeling procedure. A weak learner (LR) iteratively assigns labels to selected rejects. A strong learner (XGB) is trained on augmented labeled data and screens new applications. Performance of XGB is measured with BE after each labeling iteration.

the accuracy of the assigned labels is at risk when using a strong learner, which may be prone to overfitting the biased training data. Third, using the same confidence thresholds for labeling *good* and *bad* rejects preserves the class ratio in the augmented labeled sample. In contrast, the *bad* ratio in the population of applicants is higher than among accepts. Finally, employing commonly used stopping criteria based on the absence of examples with confident predictions may lead to exceeding the suitable number of labeling iterations, which risks overfitting and can strengthen the error propagation due to the bias.

The BASL framework addresses the limitations of self-labeling for RI with three extensions: (i) introducing a filtering stage before labeling, (ii) modifying the labeling and training regime, and (iii) introducing stopping criteria to handle sampling bias. Fig. 1 summarizes the framework. Appendix B offers pseudo-codes for all BASL stages. Before elaborating on individual BASL stages, it is important to emphasize that BASL does not aim to eliminate bias in the training data. Full debiasing seems inappropriate because the repayment behavior of rejects from distant distribution regions is hard to estimate. Thus, we augment the training set with carefully chosen rejects for which the labeling model is confident and the data distribution is not too different from accepts. By doing so, we cannot expect the augmented sample to be free of bias. Rather, we aim to reduce bias while avoiding error propagation. The trade-off between bias reduction and scorecard accuracy is a crucial element of BASL, which we examine in detail in Appendix C.4 by constructing bias-accuracy Pareto frontiers with BASL variants that label different subsets of rejects.

5.1. Filtering stage

This stage filters rejects in D^r for two reasons. First, we aim to remove rejects from the most different part of the distribution compared to D^a . Removing such cases reduces the risk of error propagation since predictions of a model trained over accepts become less reliable when the distribution of rejects shifts further from the one observed during training. We also remove rejects that are most similar to accepts because labeling these would provide little new information and might even harm performance due to introducing noise.

We estimate the similarity between rejects and accepts by scoring D^r with a novelty detection algorithm trained over D^a . To remove the rejects most and least similar to accepts, we drop examples within the top β_u and bottom β_l percentiles of the predicted similarity scores. The threshold values $\beta = (\beta_u, \beta_l)$ act as meta-parameters of the filtering

algorithm, which we implement using isolation forest, a scalable tree-based novelty detection algorithm suitable for high-dimensional feature spaces (Liu et al., 2008). Xia (2019) has also used isolation forest for filtering rejects. Interestingly, their approach differs fundamentally from BASL. They identify outliers among rejected clients and argue that these represent good applicants who were accidentally rejected. We conjecture that the fact that a rejected client differs in distribution from other rejects does not imply a repayment status and rather rely on distributional differences between accepts and rejects in BASL.

5.2. Labeling stage

After filtering, we iteratively label selected rejects. We employ distinct regimes for labeling rejects and training the resulting scorecard, and suggest scoring rejects using a learner with different inductive bias. The labeling algorithm should provide well-calibrated predictions to select the appropriate confidence thresholds. Another desideratum of the labeling algorithm is that it should be less prone to overfitting the biased training sample. Using different algorithms for RI and scoring new applications also reduces the risk of amplifying the bias of the base classifier.

We use L1-regularized logistic regression (LR) as a weak learner for labeling rejects. The L1 penalty is introduced when working with high-dimensional data with noisy features. LR is a parametric learner that outputs probabilistic predictions. Appendix B.2 shows how LR predictions are better calibrated and take extreme values less frequently compared to a strong non-parametric learner such as XGB. Another benefit of LR over tree-based models such as XGB is its ability to extrapolate outside of the range of observed feature values, which is crucial since rejects stem from a different distribution region. Relying on recursive partitioning of the training data, tree-based methods cannot extrapolate outside feature value regions observed among accepts.

In each labeling iteration, we randomly sample ρm examples from rejects. Sampling aims at preventing overfitting by examining different regions of the distribution of rejects. Assuming that the currently deployed scorecard performs better than random, we expect the *bad* rate in D^r to be higher than in D^a . Therefore, we introduce the imbalance parameter θ . We only label examples with a high confidence in the assigned labels, selecting the bottom γ percentile and the top $\gamma\theta$ percentile of the distribution of scores predicted by the weak learner. Setting $\theta > 1$ helps to append more *bad* than *good* examples, increasing the *bad* rate in the training sample to approximate the population distribution. The selected labeled rejects are removed from D^r and appended to D^a .

5.3. Training stage

At the end of each labeling iteration, we train a scoring model on D^a that was augmented with labeled rejects. The augmented sample covers a wider range of the feature space compared to the original sample of accepts. This reduces the effect of sampling bias on the trained model. Following Gunnarsson et al. (2021), we use XGB as a base classifier for the resulting scorecard to benefit from a strong base learner with high discriminative power to screen new applications.

5.4. Early stopping

The number of labeling iterations is controlled by the stopping criteria. After each iteration, we evaluate the scorecard on a holdout sample containing labeled accepts and unlabeled rejects using BE. This allows us to account for the impact of sampling bias on evaluation. If the model performance does not improve, we stop labeling and use the previous model as the final scorecard. We also specify the maximum number of labeling iterations j_{max} and terminate BASL if there are no more rejects in D^r for which predictions exceed the specified confidence thresholds.

6. Experimental setup

We perform a simulation experiment to illustrate sampling bias, demonstrate gains from our propositions, and investigate boundary conditions affecting their performance. Next, we test our methods on a real-world high-dimensional microloan data set and estimate their business value.

6.1. Synthetic data

We generate synthetic loan applications using two multivariate Gaussian mixtures:

$$\begin{cases} \mathbf{X}^g \sim \sum_{c=1}^C \delta_c \mathcal{N}_k(\mu_c^g, \Sigma_c^g) \\ \mathbf{X}^b \sim \sum_{c=1}^C \delta_c \mathcal{N}_k(\mu_c^b, \Sigma_c^b) \end{cases} \quad (1)$$

where $\mathbf{X}^g$ and $\mathbf{X}^b$ are feature matrices of *good* and *bad* applications, and δ_c , μ_c , and Σ_c are, respectively, the weight, mean vector and covariance matrix of the c th mixture component.

To mimic the scorecard-based loan approval process, we devise a simulation framework called the acceptance loop hereafter. We assume a FI approves applications using a scorecard $f_a(X)$ that predicts $\mathbf{P}(y = 1|X)$. The FI accepts applicant X if $f_a(X) \leq \tau$, where τ is a probability threshold. Suppose $D_j = \{(\mathbf{X}, y)\}$ is the batch j of i.i.d. applications with $(\mathbf{X}, y) \sim \mathbf{P}_{XY}$ where y is unknown at the time of application. Acceptance decisions partition D_j into $D_j^a = \{X_i \in \mathbf{X} | f_a(X_i) \leq \tau\}$ and $D_j^r = \{X_i \in \mathbf{X} | f_a(X_i) > \tau\}$ for accepts and rejects. Next, we reveal the labels in D_j^a and retrain the scoring model on $D^a = \bigcup_{j=1}^J D_j^a$, with J denoting the total number of batches. We use the updated model to score new applications. Over time, D^a grows in size with a bias towards accepts. We run the acceptance loop for 500 iterations, generating a new batch of applications using the same distribution parameters each time. We also draw a holdout sample from $\mathbf{P}_{XY}$ denoted as H . The sample H facilitates assessing scorecard performance and bias correction methods on unseen data representative of the borrower population. Appendix C.1 further elaborates on the simulation.

Control over the data-generating process facilitates sensitivity analysis to clarify how the loss due to bias and gains from our propositions develop with changes in the environment. Following Section 2, the sensitivity analysis comprises a gradual transition from MAR to MNAR. Other factors influencing the effectiveness of BASL include the strength of sampling bias, class imbalance, and the complexity of the classification task. Factors relevant for BE include the validation sample distribution and the class prior for labeling rejects.

Table 1

Real data summary.

Characteristic	Accepts	Rejects	Holdout
Number of applications	39,579	18,047	1,967
Number of features	2,409	2,409	2,409
Percentage of <i>bad</i> applications	39%	Unknown	66%

6.2. Real data

The real data is provided by a FinTech company, Monedo, and consists of micro-loans issued to customers in Spain. The data includes 2,409 features characterizing loan applications. The binary target variable indicates whether the customer repaid the loan (*good*) or experienced delinquency of at least three consecutive months (*bad*). The data covers 59,593 applications, of which 39,579 were accepted and 18,047 were rejected. The unlabeled rejects are leveraged by BASL, BE, and benchmarks to address the impact of sampling bias. We also have access to a labeled *unbiased* sample of 1,967 applications. Instead of scoring applicants, Monedo accepted any application for a short period. This sample, therefore, represents the through-the-door population of customers, which is not biased towards low-risk cases and includes applications the lender would normally reject. Compared to the previously accepted cases that went through scoring, the difference in distributions is clearly reflected in the percentage of *bad* applications, which reaches 66% in the unbiased sample as opposed to 39% among the accepts. The unbiased sample allows us to evaluate performance gains from our propositions under Monedo's true operating conditions.

Table 1 summarizes the data and reveals that the *bad* rate in the holdout sample is 1.7 times higher than among accepts, which evidences sampling bias. Appendix D.1 provides additional data information, including a description of the feature categories. Appendix D.2 offers an in-depth analysis of sampling bias in the real data and how it affects scorecard parameters, training, and evaluation. The corresponding results agree with those from synthetic data in Section 7.1.

6.3. Experiments

We perform two experiments to test our propositions. Experiment I tests whether BE estimates the true scorecard performance more reliably than other evaluation strategies. We consider a performance prediction setup. We split accepts into training and validation sets and apply evaluation strategies to a scorecard trained on the training data. Each strategy estimates the true performance on unbiased data from the borrower population. Ignoring sampling bias by evaluating on accepts is the baseline. DR and reweighting act as off-policy evaluation benchmarks.⁴ BE evaluates the scorecard on a merged validation set of accepts and unlabeled rejects. We assess the performance of each evaluation strategy by calculating the RMSE between the performance estimates produced over the experimental trials and the actual scorecard performance on the unbiased holdout sample.

Experiment II focuses on scorecard training. We correct the training set of accepts with one of the bias correction methods. We then train a scoring model over the corrected sample and evaluate it on a representative holdout sample. We compare BASL to several bias correction methods. Ignoring rejects serves as a baseline. Labeling rejects as *bad* and bureau score-based labeling are simple augmentation techniques popular in credit scoring. HCA and parceling represent model-based augmentation methods. The Heckman model is another benchmark suited for MNAR and established in the credit scoring literature. We also implement reweighting with cluster-based weights,

⁴ Recall that differences between the off-policy evaluation setup and credit scoring prohibit the direct application of DR. Appendix F.2 details our implementation of an adjusted DR estimator that supports credit scoring.

while the bias-removing autoencoder serves as a representation change benchmark.

The simulation study allows us to conduct the experiments within the acceptance loop and aggregate the results over 100 trials. Knowledge of the actual labels of synthetic rejects also allows us to implement an oracle model $f_o(X)$ trained on $D^a \cup D^r$. The oracle represents a scorecard that does not suffer from sampling bias and gives an upper performance bound. The real data does not support such a dynamic evaluation or an oracle solution. To improve the robustness of the results obtained on the real data, we aggregate performance over 100 values from 4 cross-validation folds times 25 bootstrap samples of the holdout set (see Appendix E for meta-parameter tuning details).

The simulation study and the real data experiment employ a set of complementary indicators to assess credit scorecards. The AUC and the Brier Score (BS) are widely used evaluation measures, which assess, respectively, the discriminatory ability of a scorecard and the degree to which its predictions are well-calibrated. Given that accepting a *bad* applicant incurs higher costs than rejecting a *good* applicant, we also consider the Partial AUC (PAUC). Summarizing the ROC curve on a limited range of cutoffs, the PAUC can account for asymmetric error costs. We compute the PAUC in the area of the ROC curve with a false negative rate in $[0, .2]$, which exemplifies a selective approval policy with low acceptance rates. Lastly, assuming FIs aim at minimizing losses while approving a specific percentage of applications, denoted as α , we consider the *bad* rate among accepts (ABR), where accepts are the top $\alpha\%$ applications with the lowest estimated PDs. Specifically, we integrate the ABR over acceptance between 20% and 40%, which reflects historical policies at Monedo. For each metric, we use Algorithm 1 to compute the Bayesian extension of that metric, which estimates performance over the unbiased holdout data H .

7. Results

7.1. Synthetic data

7.1.1. Simulation results in a MAR setup

Fig. 2 illustrates sampling bias and its adverse effects on scorecard training and evaluation. Panel (a) compares the distribution densities of one of the synthetic features in D^a , D^r and H , highlighting differences between D^a and H . The values of x_1 in $(-3, 1)$ are not observed among accepts in D^a , although the density peak in the unbiased set H is within this interval. This confirms that the data of previously accepted clients are not representative of the population of loan applicants.

Bias in the training data affects the scorecard behavior. We use a non-parametric XGB-based scorecard, which prohibits a direct inspection of the model parameters to illustrate the bias in the classifier. Regressing applicant features on the predictions of an XGB scorecard using linear regression, we obtain a surrogate model that approximates how XGB maps feature values to predictions. Panel (b) compares the coefficients of the surrogate models corresponding to the three XGB scorecards: (i) biased model $f_a(X)$ trained over D^a ; (ii) oracle model $f_o(X)$ trained over $D^a \cup D^r$; (iii) model $f_c(X)$ corrected by labeling rejects with BASL. The results indicate that sampling bias affects the coefficients of surrogate scorecards and causes them to diverge from the oracle values. BASL partly recovers this difference, bringing the coefficients closer to the oracle. The bias in model parameters translates into a difference in the scores predicted by the scorecards. As illustrated in panel (c), f_a provides more optimistic scores compared to f_o , whereas the distribution of scores produced by f_c is more in line with that of the unbiased model.

Panel (d) depicts the results of Experiment I, which examines the impact of sampling bias on scorecard evaluation. It compares the ABR of f_a on a representative holdout sample H (labeled as oracle ABR) and the estimated ABR of f_a using accepts-based vs. BE. The oracle ABR and the ABR on accepts both decrease over acceptance loop iterations due to the increasing scorecard accuracy. However, evaluating f_a on a sample

from D^a gives an overoptimistic estimate of around .0140 compared to the oracle ABR of .2095 observed on H . This gap illustrates that the accepts-based performance estimate is misleading due to sampling bias. BE provides a more reliable ABR estimate, reducing the mean RMSE between the actual and predicted ABR values from .2010 to .0929. Although BE also underestimates the oracle ABR in early iterations of the acceptance loop, it consistently predicts a substantially higher ABR than the accepts-based approach. This shows that a small number of rejected applications suffices to improve the performance estimate when using BE. After a certain number of iterations, the accumulated volume of rejected loans used by BE becomes sufficiently big and representative of the applicants' distribution to steer the Bayesian estimate to higher values, reducing the gap between the Bayesian and the oracle ABR.

Panel (e) reports the results of Experiment II, depicting the effect of sampling bias on the scorecard performance. It compares the ABR of f_a , f_o and f_c . The performance of f_a gradually improves over acceptance loop iterations due to the increasing training sample size. The ABR of f_o remains stable after 100 iterations and is consistently lower than that of f_a . The difference between f_o and f_a captures the loss due to sampling bias. The average ABR loss across 100 simulation trials is .0598. The model corrected with BASL consistently outperforms f_a starting from the first iteration of the acceptance loop. Note that f_c still suffers from the bias and does not reach the ABR of f_o . However, BASL consistently recovers about 25% of the ABR loss compared to f_a .

We observe similar performance gains from BE and BASL for other evaluation metrics (AUC, PAUC, BS) and report corresponding results for Experiment I and Experiment II in Appendix C.2 and C.3, respectively. Performing statistical hypothesis testing, we find that the observed gains are statistically significant. Friedman's rank sum tests reject the null hypothesis that our propositions perform the same as ignoring rejects in both experiments (p-values are lower than 2.2×10^{-16}). Pairwise Nemenyi post-hoc tests indicate that our propositions significantly outperform the accept-based training and evaluation at a 5% significance level.

7.1.2. Sensitivity analysis

This section examines factors that determine the effectiveness of our propositions to understand when their use is recommended and identify boundary conditions. Fig. 3 depicts loss due to bias and gains from BASL across different simulations. Panel (a) examines the influence of the magnitude of sampling bias in the training data. The bias magnitude is controlled by the acceptance rate, which varies across financial products and markets. Lower acceptance results in a greater difference between D^a and D . This raises the loss due to bias and the effectiveness of BASL in mitigating the loss. We find BASL to improve scorecards for acceptance rates below 20%, whereby this result originates from assuming a 30% *good* rate in the borrowers' population. Increasing the *good* rate would lead to a slower decay of the loss due to bias and lower performance gains from BASL. Still, the analysis implies that BASL is more helpful for financial products with low acceptance rates, such as e.g. installment loans for prime customers. The results also agree with Crook and Banasik (2004), who find a negative relationship between the acceptance rate and performance gains from reweighting-based bias correction.

Panel (b) studies the classification complexity and depicts the development of scorecard performance as a function of the feature covariance range. The elements of the feature covariance matrix are drawn randomly. A wider range of possible covariance values increases the classification complexity because loan applications of different classes overlap more frequently in the feature space. The loss due to sampling bias is consistently present across the considered complexity range. Performance gains from BASL are higher in environments with a lower classification complexity and gradually diminish in more complex environments. This is because the pseudo-labels assigned to rejects are more accurate when class separation is easier. The ability to distinguish *good* and *bad* applicants is, therefore, an important factor governing

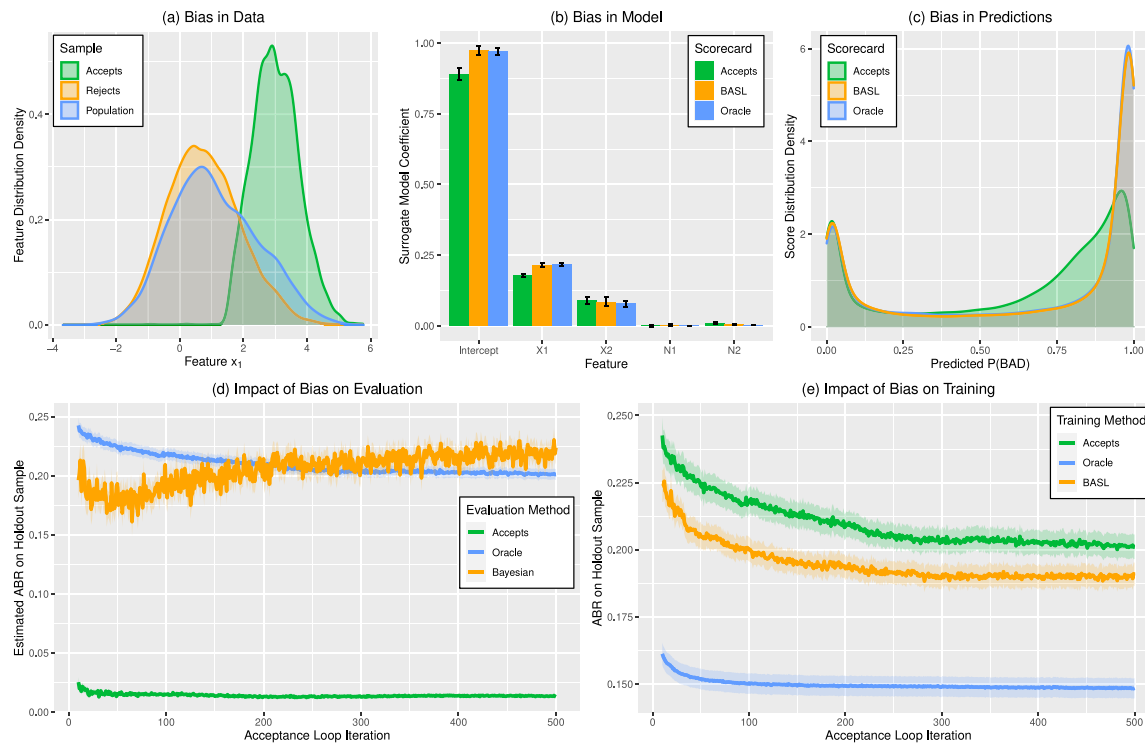


Fig. 2. Loss due to sampling bias and gains from our propositions.

Note: The figure depicts sampling bias, its effect on credit scorecards, and gains from our propositions on synthetic data. Panel (a) shows bias in the distribution of one feature, panel (b) – in scorecard coefficients based on the corresponding linear surrogate models, panel (c) – in scorecard predictions for new loan applications in the holdout sample. Panels (d) and (e) depict the impact of bias on scorecard training and evaluation in terms of the ABR.

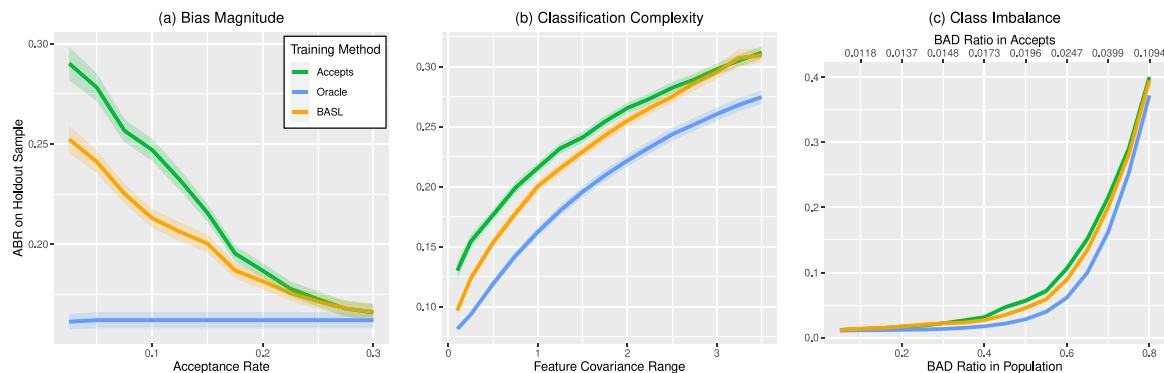


Fig. 3. Sensitivity analysis: Bias-aware self-labeling.

Note: The figure illustrates the impact of sampling bias on model training and benefits from RI with BASL as a function of three parameters: (i) bias magnitude controlled by the acceptance rate; (ii) classification task complexity controlled by the feature covariance range; (iii) class imbalance controlled by the *bad* rate in the population.

the potential of RI. In practice, observed default rates can shed light on the complexity of the loan classification task. Higher default rates evidence lower accuracy of the estimated risk scores. Low accuracy can result from an inadequate scoring model. However, financial institutions have performed risk scoring for decades, suggesting that scoring practices and processes are well-developed. A more likely explanation for high default rates is the actual complexity of the risk prediction task (e.g. high Bayes error). Recent work emphasizes the value of *alternative* data for credit risk management (e.g., Alfonso-Sánchez et al., 2024; Djeundje et al., 2021). FinTechs especially gather large, high-dimensional datasets to maximize risk prediction accuracy, as the case of Monedo illustrates. Based on results from Panel (b), a trend towards using higher dimensional (alternative) data for credit scoring could benefit BASL if the additional features raise risk predictability, rendering the classification problem less complex.

Panel (c) investigates the impact of class imbalance, which we control by the proportion of *bad* risks in the population. The results show how any *bad* rate in the population translates into class imbalance among accepts since the data is filtered by a scorecard. The loss due to bias shrinks when class imbalance becomes too strong. This comes from the ABR metric focusing on the least risky applicants, which are mostly *good* due to high imbalance. Likewise, very high bad rates in the population reduce the loss due to bias because the set of accepts comprises many bad risks from which an accepts-based scorecard can learn. BASL provides the largest gains when the imbalance level is sufficiently high so that an accepts-based model is not exposed to enough *bad* risks but is not too severe to prohibit learning from close to zero defaults. In our simulation, this range covers bad rates among accepts between 1.8 and 3%. However, a more general result of Panel (c) is that class imbalance is not a key factor explaining the effectiveness of BASL. The differences

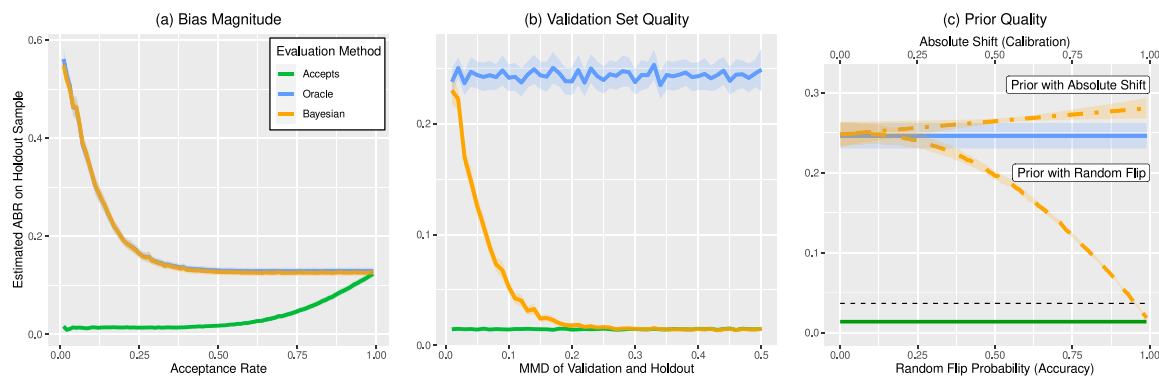


Fig. 4. Sensitivity analysis: Bayesian evaluation.

Note: The figure illustrates the impact of sampling bias on scorecard evaluation and benefits from BE as a function of three parameters: (i) bias magnitude controlled by the acceptance rate; (ii) validation set quality controlled by the accepts rate and measured by the MMD; (iii) prior quality controlled by adding noise as a label flip or level shift. A dashed black line gives the ABR predicted by BE when using the empirical *bad* rate on accepts as a prior.

in ABR across the scorecards are relatively small across all imbalance ratios considered in the simulation.

Turning to BE, panel (a) of Fig. 4 examines the effect of the acceptance rate on scorecard evaluation. To isolate this effect, we assume a perfect prior when calculating the Bayesian extension of the ABR. Under this assumption, BE accurately estimates performance across all acceptance rates. Similar to BASL, potential gains from BE are higher at lower acceptance, as the inconsistency between the performance on accepts versus a representative sample becomes stronger.

Calculating the Bayesian extension requires a validation sample of labeled accepts and unlabeled rejects. Panel (b) studies how the quality of this sample affects evaluation. We assess sample quality using the maximum mean discrepancy metric (MMD, Borgwardt et al., 2006), which measures the similarity of the feature distribution in the validation set and the unbiased holdout set. The results reinforce accept-based evaluation to underestimate error rates substantially. To predict scorecard performance accurately, BE requires validation data that matches the target distribution in the holdout set. To ensure this, the validation sample should include accepts and rejects from the same period and match the accept/reject ratio in the holdout sample.

Panel (c) focuses on the class prior for labeling rejects. Reducing the quality of the estimate of the prior decreases the accuracy of BE. The quality reduction can stem from diluting the calibration quality of the prior, which we simulate by injecting random flips or level shifts, respectively. BE provides more accurate performance estimates than accepts-based evaluation even under severely corrupted priors. This implies that using imperfect scores (e.g., from a currently used scorecard) as a prior for the labels of rejects still produces a better result than ignoring rejects during evaluation.

7.1.3. Missingness type

The previous results are based on data following an MAR process, where all applicants' features are revealed to the scorecard. Next, we introduce MNAR by generating data with three explanatory features and using only the first two in the scorecard, creating an omitted variable bias. On each iteration of the acceptance loop, we overwrite a percentage of the scorecard-based acceptance decisions by replacing applicants with the lowest values of the hidden feature. This allows us to gradually increase the strength of the MNAR process in the data.

Fig. 5 illustrates the loss due to sampling bias and the performance of our propositions depending on the overwriting strength. We implement two established bias correction benchmarks: BASL is compared to the Heckman model, which is developed for MNAR settings, whereas BE is compared to DR evaluation. We observe adverse effects of sampling bias on the training and evaluation of scorecards at any level of overwriting. This confirms that bias correction is required regardless of whether the data exhibits MAR or MNAR. In both extremes with no

or full overwriting, loss in the ABR exceeds .05 during training. The impact on evaluation is stronger with a mean difference of .21 between the ABR estimated on accepts and the ABR on holdout applications.

Our propositions consistently outperform accepts-based training and evaluation. Panel (a) reveals a poor performance of the Heckman model under MAR. We explain this by a high correlation between the two variables considered in the model: outcome (i.e., whether the applicant is *bad*) and selection (i.e., whether the applicant was accepted), which was also observed in previous studies (Chen & Astebro, 2001). The correlation is high when scorecards are accurate. Traditional banks that focus on prime customers often obtain low default rates, which indicates the high accuracy of their scorecards. Increasing the overwriting percentage reduces the target-missingness correlation and strengthens the magnitude of MNAR, facilitating a superior performance of the Heckman model. Heckman outperforms BASL when more than 20% of the applications undergo overwriting. In practice, such a large degree of manual intervention seems implausible for data-driven FIs, which increasingly rely on the automation of approval decisions but might be realistic for some banks, which would then be better served by using a Heckman-style model. However, since the type of missingness is difficult to identify in practice, the consistent superiority of BASL over the accepts-based benchmark leads us to conclude that FIs adopting BASL will not run the risk of impeding scorecard performance; which they might when using the Heckman model.

Considering model evaluation, we observe the accuracy of the ABR estimates from BE to decrease in the MNAR setting. However, they are much closer to the true scorecard performance than estimates coming from accepts-based evaluation and DR across all setups. DR improves over ignoring rejects but still provides ABR estimates that are, on average, more than .10 off the actual model performance. This large gap can be explained by the adjustments required to apply DR in credit scoring, which we discuss in detail in Appendix F.2.

7.2. Real data

7.2.1. Experiment I: Evaluation with the Bayesian framework

Table 2 depicts the accuracy of model evaluation strategies ordered by their average rank from best to worst approach. We compute the RMSE between the scorecard performance estimates produced by each strategy and the actual performance on the holdout sample representing the true borrower population. In line with the synthetic data results, we observe a relatively high RMSE when ignoring rejects due to sampling bias. Overoptimistic scorecard performance estimates from the accepts-based evaluation lead to sub-optimal decisions. Expecting a certain default rate upon model deployment, an FI could face losses and liquidity problems when encountering a substantially higher default rate among approved loans.

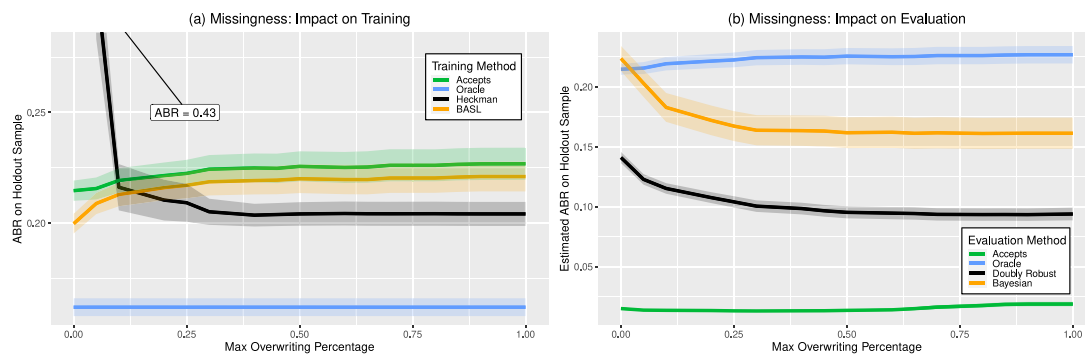


Fig. 5. Sensitivity analysis: Missingness type.

Note: The figure depicts the relationship between overwriting and the impact of sampling bias on scorecard training and evaluation, benefits from our propositions (BASL and BE) and benchmarks (Heckman model and DR).

Table 2
Scorecard evaluation: comparing performance of bias correction methods.

Evaluation method	Rank	AUC	BS	PAUC	ABR
Bayesian evaluation	1.06	.0111 (.1158)	.0073 (.0213)	.0351 (.0628)	.0130 (.0331)
Ignore rejects	2.46	.1234 (.0309)	.0306 (.0034)	.0983 (.0246)	.0356 (.0603)
Reweighting	2.49	.1277 (.0601)	.0348 (.0054)	.0826 (.3058)	.0315 (.0903)
Doubly robust	–	–	.0506 (.0050)	–	.1167 (.0216)

Values indicate RMSE between the actual scorecard performance on the holdout sample and estimates of that performance using an evaluation method. Variance of the performance estimates $\times 10^{-5}$ in parentheses.

Weighted validation improves the accuracy of scorecard performance estimates for two evaluation metrics. Overall, reweighting performs marginally worse than accepts-based evaluation, achieving an average rank of 2.49 compared to 2.46. At the same time, reweighting outperforms accepts-based evaluation in the metrics that account for asymmetric error costs, PAUC, and ABR, which are most important for decision-makers. DR demonstrates a poor RMSE for the two supported metrics, BS and ABR. This can be attributed to the high difficulty of reward prediction in a high-dimensional environment and the limitations of DR when applied to credit scoring. Poor performance in the BS and ABR while lacking support for rank-based indicators such as the AUC and PAUC make DR an inappropriate evaluation method for the considered data set.

BE provides the most accurate scorecard performance estimates across all evaluation metrics and achieves an average rank of 1.06. This implies that BE produces the most reliable predictions of model performance on new loan applications, helping decision-makers to anticipate the accuracy of a scorecard and judge its (business) value ex-ante. Appendix D.3 augments Table 2 with statistical testing results. Pairwise Nemenyi post-hoc tests indicate that performance estimates obtained with BE are significantly better than those obtained with benchmark strategies at a 5% level.

7.2.2. Experiment II: Reject inference with BASL

Table 3 shows the performance of bias correction methods ordered by their average rank from best to worst approach. Several methods perform worse than ignoring rejects. Labeling rejects as *bad* performs the worst. Given a historical acceptance rate of 20–40% at Monedo, the underlying assumption of all rejects being *bad* is too strong for this data. The bias-removing autoencoder also performs poorly. As discussed in Appendix F.3, due to a large number of features and a broad set of meta-parameters, the reconstruction error of the autoencoder remains high even after much tuning. This evidences the difficulty of using an autoencoder in high-dimensional settings.

The bivariate probit model improves on the previous benchmarks but performs worse than ignoring rejects. Its poor accuracy can be attributed to two reasons. First, a parametric probit model faces difficulties in handling high-dimensional and noisy data. To address this, we explore multiple variants of Heckman-style models described in

Appendix F.4 that use different feature subsets, estimation procedures, and base classifiers. The best-performing two-step model presented in Table 3 uses XGB to model the outcome process and outperforms bivariate probit. Second, in line with the synthetic data results, the Heckman model performs poorly when the outcome and selection equations are highly correlated, which is typical under MAR or when the previous scorecard is accurate. Although it is infeasible to estimate the strength of the MNAR process on the real data, the poor performance of Heckman suggests that the missingness type is more geared towards MAR. A fully automated loan approval pipeline at Monedo further supports this view.

Considering model-based augmentation techniques, HCA performs on par with a two-step Heckman model, improving on ignoring rejects only in the AUC, whereas parceling performs better in three evaluation measures. The better performance of parceling can be explained by introducing randomness in the labeling stage, which helps reduce error propagation. Reweighting outperforms other benchmarks in the AUC, PAUC and ABR. However, it performs worse than parceling in terms of BS, indicating poor calibration of the resulting scorecard. This translates to a marginally higher overall rank of reweighting compared to parceling. Appendix F.1 discusses the performance of different reweighting variants, whereas Table 3 only includes the best-performing specification.

Independent of the performance indicator, BASL performs best and achieves the lowest average rank of 1.72. Compared to reweighting, the closest competitor in the cost-sensitive metrics, the PAUC and ABR of the scorecard after bias correction with BASL increase by .0114 and .0135, respectively. Gains from BASL are statistically significant: Nemenyi post-hoc tests indicate that BASL significantly outperforms all benchmarks at a 5% level in the AUC, PAUC, and ABR. Table D.6 in Appendix D provides ablations, which examine incremental performance gains from different BASL stages. While all stages add value, the largest gains stem from the filtering step.

Experiment I and II confirm that our propositions outperform previous bias correction methods on the employed high-dimensional FinTech data set. To test external validity, we repeat experiments using gradually smaller subsets of features, simulating the transition from FinTechs to traditional banks, which tend to use parsimonious scorecards. The results suggest that BASL and BE consistently reduce sampling bias across all settings, with larger gains observed on higher-dimensional data. The corresponding results are available in Appendix D.5.

Table 3

Scorecard training: comparing performance of bias correction methods.

Training method	Rank	AUC	BS	PAUC	ABR
Bias-aware self-labeling	1.72	.8166 (.0007)	.1761 (.0003)	.7075 (.0011)	.2211 (.0012)
Parceling	3.83	.8038 (.0011)	.1804 (.0004)	.6885 (.0011)	.2396 (.0019)
Reweighting	3.91	.8040 (.0005)	.1840 (.0002)	.6961 (.0009)	.2346 (.0015)
Two-step Heckman model	4.03	.8034 (.0011)	.1828 (.0004)	.6934 (.0014)	.2381 (.0019)
Ignore rejects	4.12	.7984 (.0010)	.1819 (.0004)	.6919 (.0010)	.2388 (.0019)
Hard cutoff augmentation	4.84	.8033 (.0010)	.1830 (.0006)	.6790 (.0011)	.2458 (.0021)
Bureau score based labels	5.69	.7978 (.0009)	.1860 (.0003)	.6783 (.0010)	.2514 (.0021)
Bivariate probit	8.53	.7444 (.0011)	.2124 (.0006)	.6397 (.0010)	.3018 (.0013)
Bias-removing autoencoder	8.77	.7304 (.0011)	.2161 (.0004)	.6376 (.0019)	.3061 (.0036)
Label all as bad	9.56	.6676 (.0014)	.2347 (.0006)	.6384 (.0010)	.3141 (.0022)

Values in parentheses represent standard errors across 100 trials of 4-fold cross-validation $\times$ 25 bootstrap samples.

7.2.3. Business impact analysis

Estimating monetary gains from BASL and BE requires knowledge of key loan parameters. We consider the microloan market and draw loan amounts and interest rates from Gaussian distributions. In line with statistics from the US consumer loan market ([The Pew Charitable Trusts, 2016](#)), we assume a mean principal $A = \$375$ and a mean interest $i = 17.33\%$.

In the event of default occurring with a probability PD, a financial institution recovers $A \times (1 + i) \times (1 - \text{LGD})$. If there is no default, the expected revenue is $A \times (1 + i)$. We approximate the loan-level PD by computing the ABR of the scorecard within a specified acceptance range. Using the modeling pipeline described in Section 6, we obtain 100 ABR estimates for each scorecard. Given these 100 estimates, Eq. (2) yields an estimate of the average profit per loan:

$$\pi = \frac{1}{100} \sum_{j=1}^{100} [\text{PD}_j \times A \times (1 + i) \times (1 - \text{LGD}) + (1 - \text{PD}_j) \times A \times (1 + i) - A] \quad (2)$$

We aggregate the average profit per loan over 10,000 trials, drawing A and i from Gaussian distributions and varying LGD from 0 to 1. By subtracting the profit of each bias correction method from the profit of an accept-based scorecard, we obtain the incremental profit compared to ignoring rejects (and sampling bias). Finally, we compute the expected margin (i.e., the expected return per dollar issued) by dividing the incremental profit by the average loan amount. Following the historical approval rates at Monedo, we vary the acceptance rates in 20–40%.

BE improves the scorecard evaluation accuracy. To estimate gains from better evaluation, we consider a scenario in which a FI adjusts acceptance rates to maximize profit. We use BE and benchmarks to estimate the expected ABR of a scorecard across different acceptance rates in 10–50%, extending the range beyond the historical approval rates at Monedo. We use each evaluation strategy to select the acceptance rate that maximizes profit. Finally, we compare incremental profit under the acceptance rate selected based on BE, and the one based on reweighting.

Fig. 6 illustrates the expected incremental profit as a function of the LGD. In both settings, ignoring sampling bias impacts the profit of a financial institution. BASL increases the expected return per dollar issued by up to 2.07 percentage points compared to ignoring rejects and up to 1.58 compared to reweighting, whereby gains increase with the LGD (as higher LGDs raise the cost of accepting a *bad* customer). BE increases the expected return per dollar issued by up to 5.67 percentage points compared to both benchmarks. The gains are highest for LGD between 40% and 80% and diminish at very low or very high loss rates. At LGD below 30%, all evaluation strategies start recommending the maximal acceptance rate, as the cost of accepting a *bad* customer becomes very small. Similarly, at LGD close to 100% all evaluation strategies converge to recommending the minimal acceptance due to a high cost of accepting a *bad* customer.

The business impact analysis demonstrates consistent monetary gains from the proposed bias correction methods. Assuming the average loan amount of \$375 and the mean LGD of 50%, the incremental

profit per loan is \$2.97 for BASL and \$29.27 for BE. This shows that correcting the impact of sampling bias on evaluation promises much larger financial rewards.

8. Discussion

Our propositions target the key steps in an ML workflow: training and evaluation. For scorecard training, simulation results suggest that sampling bias deteriorates scorecard performance, assessed in terms of the ABR, by, on average, six percentage points. Using BASL, we can recover about 25 percent of the loss due to bias. Results from the real-world data support the simulation and show BASL to consistently outperform several benchmarks by a 1.6 and up to 30 percent margin, depending on the accuracy indicator. According to the business impact analysis, these gains in accuracy increase the expected return per dollar from 1.5 to 2 percentage points compared to benchmarks and the ignoring rejects baseline.

The sensitivity analysis uncovers when BASL improves scorecard training and facilitates some policy prescriptions. Observing gains from BASL to increase with decreasing acceptance rates and lower rates of good clients among applicants, we recommend using BASL for prime and/or non-collateralized financial products that FIs offer selectively. Building on supervised ML, BASL is not immune to the class imbalance problem. More specifically, neither settings with very high imbalance nor settings with even class distributions suggest the application of BASL. BASL requires some defaults in the training data to infer a relationship between applicant characteristics and repayment behavior. On the other hand, a training set with many defaults shows less sampling bias, which reduces the need to use BASL.

Using the real data, we study how the impact of sampling bias on scorecard accuracy varies with dimensionality, comparing parsimonious scorecards with a handful of characteristics to scorecards with thousands of features. We find BASL to be most effective in high-dimensional environments. In lower-dimensional settings, improvements are modest but consistent. These findings speak to the trend towards using novel data sources for credit scoring (e.g., [Djeundje et al., 2021](#)). Higher dimensionality may amplify sampling bias. Likewise, we conjecture that data-driven FIs, which use sophisticated scoring methodologies, will benefit from BASL. We also expect FIs that have already deployed advanced scoring systems to be particularly alert about predictive performance, which would raise the acceptance of scoring innovations like BASL among managers and users.

The missing data mechanism has crucial implications for RI ([Banasik & Crook, 2007](#)). We elaborate on drivers for MAR and MNAR in credit scoring and examine the performance of BASL when gradually increasing the strength of confounding in the simulation. The experiment finds that BASL outperforms the well-known Heckman model, designed for MNAR, across a range of plausible operating conditions. Heckmann-style models yield extremely poor risk predictions when the missingness mechanism leans toward MAR. The Heckman benchmark may outperform BASL by a small margin under extreme conditions, for example, assuming that credit analysts manually overwrite more

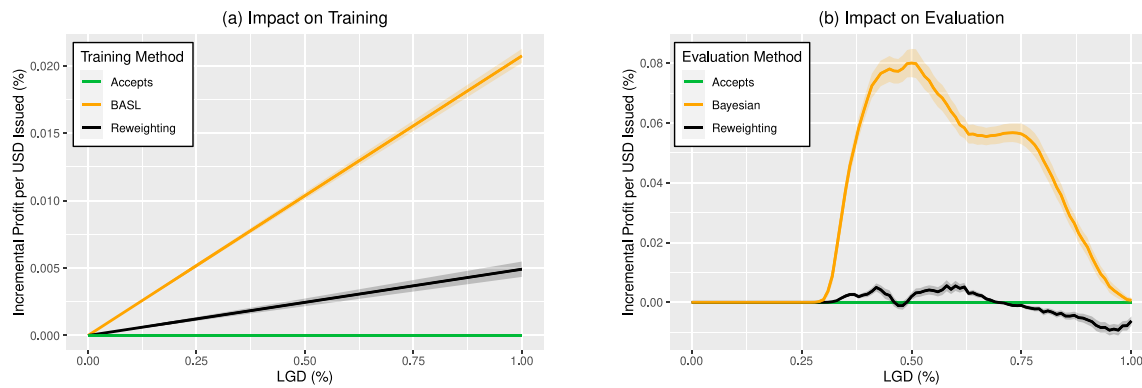


Fig. 6. Business impact analysis.

Note: The figure shows the expected incremental profit per dollar issued of our propositions and reweighting relative to a scorecard that ignores sampling bias. Panel (a) depicts gains from BASL on the scorecard training stage; panel (b) shows gains from BE in a policy change scenario of selecting a new acceptance rate.

than 25% of scorecard recommendations. Overall, our results provide evidence in favor of using BASL to address sampling bias. It promises competitive and robust performance independent of the missingness form and avoids the risk of catastrophic predictions. Alternatives like Heckman are preferable if analysts have reasons to suspect their scoring practices cause available data to exhibit strong MNAR.

Another insight from our analysis concerns the trade-off between sampling bias mitigation and predictive accuracy. The missing data literature and much prior work on RI advocate the removal of bias through estimating (missing) labels for all rejected clients. We observe a trade-off between debiasing and scorecard performance. Examining the corresponding Pareto frontiers, we find that labeling a small, carefully selected set of rejects improves scorecard performance. In contrast, noise in the estimated labels causes a fully debiased data set to yield weaker scorecards. The design of BASL accounts for this trade-off, which contributes to outperforming previous RI methods.

Turning to model evaluation, we find that the adverse impact of sampling bias is greater than in training and that BE facilitates substantial improvements over accept-based evaluation. In the simulation, we find the latter to severely underestimate default risks, predicting the ABR of a scorecard to be, on average, 0.01 compared to a true value of, on average, 0.21. BE corrects deviations between true and predicted ABR by about 54%. It also shows an appealing tendency that larger errors, if they occur, come from underestimating performance, which implies more prudent risk estimates. Results from the real-data experiment confirm accuracy gains from BE. For our micro-loan setting, improvements in predicting ABR are close to 60 percent over the strongest competitor. We also observe substantial accuracy gains for other performance indicators.

Appraising the business value of these improvements is nontrivial because scorecard performance prediction is a sparsely researched field, and reference practices are lacking. We consider a policy optimization setting in which competing approaches forecast lending profits across alternative acceptance policies. For our micro-lending context, we observe BE to raise the expected return per dollar issued by up to 5.7 percent (depending on LGD). Assuming an average loan amount of 375\$ and an average LGD of 50%, which reflect the business history of Monedo, BE increases the incremental profit per loan by 29.27\$, which equates to an eight percent improvement in profit.

Sensitivity analysis secures the efficacy of BE. All experimental conditions (e.g., varying acceptance rates, different feature sets, missingness mechanism) provide strong evidence in favor of BE. The corresponding forecasts are consistently much more accurate than alternative approaches to scorecard evaluation, facilitating a clear recommendation. Whenever FIs have access to the data of rejected clients, they should leverage this data for scorecard evaluation using BE.

Notably, this recommendation extends to the challenging MNAR case, although BE uses biased scores to infer the labels of rejects. The

sensitivity analysis supports this approach. Using imperfect scores from the current – biased – scorecard as a prior for the labels of rejects produces better results than ignoring rejects during evaluation, even if the scores embody a lot of noise. This finding indicates that BE is effective and easy to use. FIs can use the original score of an applicant or their current scorecard to set the prior, which is a key ingredient of BE.

9. Conclusion

Much research has shown the potential of ML-based scoring models to support decision-making in the credit business in static (e.g., Bravo et al., 2013; Lessmann et al., 2015) and dynamic (e.g., Alfonso-Sánchez et al., 2024; Zandi et al., 2025) settings. Considering loan approval decisions, the paper illustrates the adverse effects of sampling bias on the training and evaluation of scoring models and proposes methodologies to address these effects. BASL targets scorecard training and mitigates sampling bias by augmenting the training data with a carefully selected set of rejects with an inferred label. Concentrating on evaluation, BE also leverages unlabeled data of rejected clients by assessing scorecards on a more representative holdout test set drawn from labeled accepts and rejects with sampled labels.

Using real-world lending data, including an unbiased sample of randomly accepted loans, we confirm the effectiveness of BASL and BE and demonstrate their superiority over various benchmarks. A business impact analysis evidences sizeable improvements in profit, especially for BE. Results from a simulation study augment those findings by identifying boundary conditions that determine the effectiveness of our propositions.

Our study has several implications for credit scoring research and practice. We show that widely used evaluation regimes are vulnerable to sampling bias and give misleading advice. BE anticipates the loss in accuracy due to bias and facilitates quantifying this effect before scorecard deployment. This forward-looking performance evaluation facilitates improving operational acceptance decisions to raise profitability and can help to improve risk management practices in the credit industry.

The paper also confirms that training on biased data impairs scorecards. FIs can use BASL with any supervised learning algorithm to reduce loss in model performance. Doubt as to whether RI is worthwhile prevails in the literature (e.g., Chen & Astebro, 2001). The paper speaks to this debate by reporting positive results from an unbiased evaluation sample. RI is a challenging problem. Financial rewards will not be excessive. However, the specific engineering of BASL facilitated consistent and material gains in this study. Improvements of the magnitude observed here in a core business process may be a deciding factor in highly competitive lending markets.

A caveat of BASL and BE is that they require access to unlabeled data of rejected clients. Although this requirement is omnipresent in the RI literature, its implications are nontrivial. FIs need to store data on rejected applicants, posing thorny questions about data privacy and consumer protection. Balancing the interests of lenders to gather more data for improving processes such as loan approval and the interests of consumers for protection against privacy infringement is a significant challenge in the digital economy. Quantifying the value of a specific type of data in a specific business context, the paper contributes a humble piece of evidence to this societal debate, which may inform bank governance and regulatory authorities.

CRedit authorship contribution statement

Nikita Kozodoi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stefan Lessmann:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Morteza Alamgir:** Writing – review & editing, Supervision, Resources, Conceptualization. **Luis Moreira-Matias:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Konstantinos Papakonstantinou:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Acknowledgments

The authors are grateful for many fruitful discussions with Monedo's data science team. Yiannis Gatsoulis's insightful comments were instrumental to this research. Stefan Lessmann acknowledges funding by the project "AI4EFin Artificial Intelligence for Energy Finance" under Romania's National Recovery and Resilience Plan PNRR-III-C9-2022-I8, Contract CF162/15.11.2022.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejor.2025.01.040>.

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